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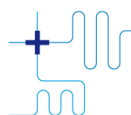
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A molecular dynamics study of chemistry-inspired nanoporous aluminum structures

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ABSTRACT

Nanoporous metal structures are of high interest in both academic research and industry due to their tunable mechanical properties and light weight. However, understanding the mechanical properties of these materials is still a challenge. Existing research has demonstrated that among many parameters (relative density, ligament diameter, specific surface area, etc.), topology plays a key role in determining the structural properties. Recent reports have focused mostly on randomly generated Al or Au structures and on several common topologies such as FCC, BCC, and gyroid. This study focuses on an alternative set of topologies reported in the context of the reticular chemistry of nanoporous (non-metallic) crystalline materials. Specifically, by performing a series of molecular dynamics simulations, 22 topologies were examined in terms of yield stress and Young's modulus, while comparing them to stochastic nanoporous aluminum structures. The results indicate that structures constructed with predefined topologies have superior Young's modulus and yield strength. Moreover, distinct scaling laws of mechanical properties are observed for different topologies, revealing the relationship between topology and mechanical behavior in nanoporous materials.

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I. INTRODUCTION

Porous metals, with pore diameters ranging from nano- to millimeter scale, are increasingly important in technology. Their high surface-to-volume ratio together with optimizable mechanical properties make these materials promising candidates in various emerging applications such as metallic scaffolds for load-bearing bones,^{1,2} lightweight structures for transport, aerospace industry,³ electrodes for electrochemical energy storage⁴ devices, and more. Designing a porous structure with the specific mechanical properties and behavior is not a trivial task as there are a number of variables affecting these properties. Such factors are topology,^{5,6} choice of material,^{5,6} fabrication method,^{5,7} and post-processing treatment.⁸

One of the main characteristics of nanoporous materials is their solid fraction $\varphi = \rho/\rho_s$, where ρ and ρ_s are the mass densities of the porous material and of the solid phase, respectively. While nanoporous metals typically have a solid fraction greater than 0.25, the Gibson–Ashby model⁹ for low-density cellular open foams is frequently applied to analyze these materials. In this case, the

scaling of the macroscopic values of Young's modulus, denoted as E , and yield strength, denoted as σ_y , are determined based on the solid fraction,

$$\frac{E}{E_s} = C_E \varphi^{n_E}, \quad (1)$$

$$\frac{\sigma_y}{\sigma_{ys}} = C_\sigma \varphi^{n_\sigma}, \quad (2)$$

where E_s and σ_{ys} are the modulus and the yield stress of the solid phase, C_E and C_σ are dimensionless prefactors dependent on topology, and n_E and n_σ are powers dependent on deformation behavior.

Although the Gibson–Ashby model is widely used in porous materials featuring microstructural dimensions of micrometers or more, its accuracy and validity could significantly suffer when applied to nanoporous materials, mostly due to their substantial

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surface-to-volume ratio. Therefore, the conventional Gibson–Ashby model should be modified to account for the deformation mechanism of nanoporous structures. Sun *et al.*¹⁰ showed that in the case of nanoporous gold samples characterized by an average ligament size of about 100 nm, the predicted yield strength according to the Gibson–Ashby model may be lower by the order of 10–100. The authors investigated the mechanism of uniaxial tensile deformation behavior of nanoporous gold with Molecular Dynamics (MD) simulations and proposed the modified scaling law for effective Young's modulus,

$$\frac{E}{E_s} = C_{E,b}\varphi^2 + C_{E,t}\varphi, \quad (3)$$

where $C_{E,b}$ and $C_{E,t}$ are dimensionless constants accounting for bending and tensile deformation of ligaments. The authors concluded that conventional open-foam materials with a low relative density follow the classical Gibson–Ashby model, while nanoporous gold structures from their study fitted to $C_{E,b} = 0.14$ and $C_{E,t} = 0.136$. It was also discovered that in case when the axial yielding of ligaments is dominating, the G-A model transforms to

$$\frac{\sigma_y}{\sigma_{ys}} = C_\sigma\varphi, \quad (4)$$

where C_σ is also constant.

In the work of Winter *et al.*,¹¹ the authors studied another common nanoporous material—aluminum. Their results of MD compression simulations showed that neither the Gibson–Ashby model nor the scaling laws for nanoporous gold proposed by Sun *et al.* are suitable for aluminum structures. From the fitting curve of MD results, it was observed that Young's modulus does not change according to the Gibson–Ashby model. Therefore, Winter *et al.* proposed modified scaling laws suitable for nanoporous aluminum. The same results were observed regarding the yield stress—the GA model does not capture the steep change of yield stress in nanoporous Al.

Despite the large number of publications involving nanoporous structures, the exact effect of topology is still far from being fully understood. However, it has become evident that porosity of a structure is not one-to-one correlated with mechanical properties. For instance, in the work of Mangipudi *et al.*,¹² researchers conducted detailed morphological and topological characterization of four types of structures: two nanoporous Au samples, spinodal structure, and gyroid structure. These structures with equal porosity were tested with Finite Element (FE) simulations to obtain mechanical properties. The stress–strain response of structures was very distinct despite the same void fraction. From these results, it is obvious that mechanical properties do not simply depend on relative density, but they are highly affected by the structure's topology. Other important conclusions of this study were that mechanical properties are more sensitive to topology rather than morphology of structures. The effect of topology was captured by modifying Gibson–Ashby scaling laws where geometric prefactors were linearly linked to the scaled genus density. One more finding of this work is that the scaled genus density of structures of the same porosity is highly dependent on the topology. Topological genus is

the maximum number of cuts that can be made for a structure before it loses integrity; usually, to compare the genus of different structures, it is scaled by volume and specific surface.

Substantial work regarding topology effects was done by Huber,¹³ it covers four different topologies: Gibson–Ashby, diamond, cubic and BCC, which are 3-, 4-, 6-, and 8-coordinated, respectively. These topologies were modified by randomly cutting part of ligaments and the resulting decline of normalized Young's modulus and yield stress was recorded. The author observed that fit curves for Young's modulus are almost identical to ones constructed for yield stress, leading to the conclusion that these two properties equally depend on the cut fraction. Furthermore, Saane *et al.*¹⁴ did a study on ordered (gyroid and cubic) and disordered (nanoporous gold) structures in terms of their charge-induced deformation and plastic strength. Despite the topological diversity of studied structures, the actuation strains exhibited similar behavior, following the scaling law. Additionally, the study reports that irrespective of structural variations, the charge-induced strains remain considerably below the plastic limit. In the research done by Li *et al.*¹⁵ researchers studied three types of lattices (cubic, gyroid, and diamond) and compared them to stochastic bi-continuous structures. The results of this latter work present scaling laws of mechanical properties depending on relative density.

In this work, we investigate the effects of topology on the yield stress and Young's modulus of nanoporous aluminum structures through a series of molecular dynamics simulations. Unlike the previous studies featuring common topologies (BCC, FCC, gyroid, etc.), we expand the set of investigated topologies to include chemistry-inspired structures. These novel topologies were obtained from Reticular Chemistry Structure Resource (RCSR),¹⁶ which is a database of structures reported in crystalline nanoporous materials such as Metal-Organic Frameworks. These novel topologies are then compared to the mechanical properties of stochastic nanoporous aluminum structures, and the effects of topology are discussed.

II. METHODOLOGY

A. Topologically defined structures

Initial data for generating topologically defined structures were obtained from RCSR database,¹⁶ which contains around 2700 topologies with different connectivity. Most of the four-connected topologies in RCSR are derived from well-studied crystal formations like zeolites, silica forms (like coesite and keatite), framework silicates like feldspar, and different polymorphs of ice (ice IV, etc.). Topologies with other coordination originate from a wide range of crystal structures. For instance, three- and six-connected nets are found in structures like rutile and anatase, etc. Many networks have been discovered in Metal-Organic Frameworks (MOFs) and similar materials. Although there is no strict rule for including a net in the database, simple structures with high symmetry are usually preferred as they have few distinct vertex types.

The following steps were done to perform simulations with desired topologies: (1) identification of three-, four-, and six-connected topologies with the RCSR search engine; (2) assembling a periodic unit cell for every found topology from the previous step

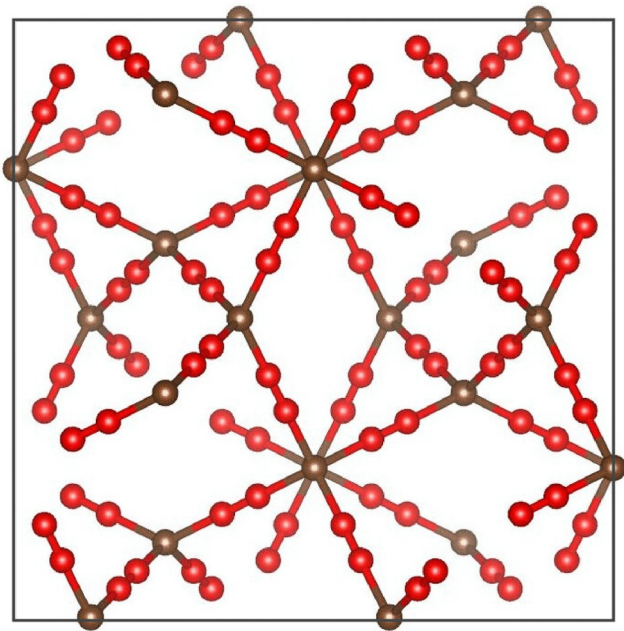


FIG. 1. *lcs* topology as a framework of C and O atoms.

with the Zeo++ tool; (3) selection of topologies, which satisfy the criteria of struts spatial orientation; (4) construction of triangulated surface in *.stl* format for every selected topology; and (5) generating input data for simulations was done by converting *.stl* file to binary 3D array and filling it with Al atoms. These steps and tools will be covered in more detail in Secs. II B and II C.

Zeo++¹⁷ is an open-source tool for performing high-throughput geometry-based analysis of porous materials and their voids. Zeo++ was developed to analyze materials such as zeolites, Metal-Organic frameworks, and diverse porous crystalline architectures. Zeo++ offers a wide range of capabilities, from which “framework builder” module was used to combine molecular building

blocks and calculate the coordinates of all the vertices as an input for *.stl* mesh generator.

To construct a framework of specified topology, Zeo++ was provided with building blocks for vertices and edges. For simplicity, carbon atoms were chosen to be placed at vertices and oxygen atoms at the edges. Figure 1 presents the *lcs* topology as a framework constructed with Zeo++. Single carbon atoms (brown) are placed at each vertex and oxygen linkers (red) are placed at each edge to connect these vertices. The visualization of the frameworks was done in VESTA.¹⁸

From all the generated frameworks, a subset of topologies was selected to be characterized with simulations. The following selection strategy was implemented. First, it was reported in the literature¹² that the ligament network of nanoporous metals is dominated by three-connected nodes (more than 80%), followed by four-connected nodes (less than 20%), with five- and six-connected nodes corresponding to the rest. To reflect this connectivity and to stay relevant with stochastic structures, we focused on three-, four-, and six-connected RCSR nets. Screening the database, we have found 254 three-connected, 613 four-connected, and 284 six-connected topologies. Second, as LAMMPS can only apply uniaxial compression along the three principal axes, we computed a load-bearing descriptor for each topology. It captures the sum of ligament projections onto the compression axis over a set of rotated copies, ensuring that the original version has the highest alignment. Figure 2 gives an example of how this descriptor changes in *dia* topology. As the structure rotates, its ligaments become more horizontally oriented, which is reflected in the decreasing sum of projections. This descriptor was motivated by the observation that struts aligned with the direction of compression contribute more to mechanical strength than perpendicular or tilted ones.¹⁹ Combining these two criteria resulted in a subset of 22 topologies, which are comparable to stochastic NPA structures and, at the same time, suitable for computationally expensive large-scale MD simulations.

The list of selected topologies is as follows:

- **Three-connected:** ana, lcs-f, pbg, pbz-l, xaa, srd, ttc, ooe, mct, kgq;
- **Four-connected:** dia, lcs, urh, gie, ukx, bcv;
- **Six-connected:** sxg, sxn, lcy-e, smq, sxe, sxf.

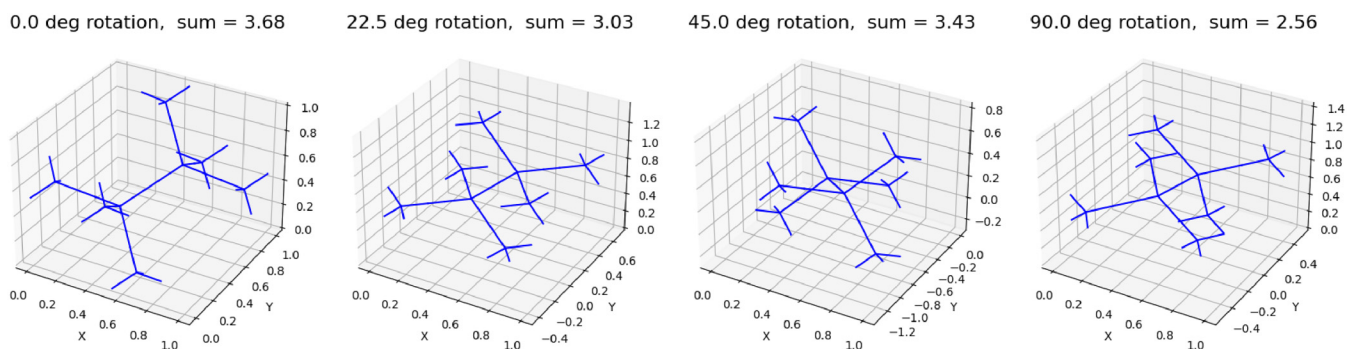


FIG. 2. Rotation of *dia* structure with the corresponding sums of struts projections to the z axis.

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For each of the 22 selected topologies, 5 copies with 5 different as-designed solid fraction values were generated 0.1, 0.2, 0.3, 0.4, and 0.5. These values were later recalculated from *.imp* files to obtain the as-built solid fraction. Performing five simulation per topology with varying solid fraction ensured that scaling laws have enough data point for fitting and that there are no deviations at certain solid fraction values. In addition to varying solid fraction, each copy was generated with a different ligament diameter 5, 7.5, and 10 nm. The initial selection of 5 nm diameter was guided by previous work on NP metals,¹⁹ where the authors concluded that the diameter of 5–6 nm is sufficient for ligaments to be free from defects after relaxation. At the same time, to observe possible deviations of scaling laws with changes in diameter, we included 7.5 and 10 nm structures in the simulations. This increased the total number of topologically defined structures to 330 (22 topologies $\times$ 3 diameters $\times$ 5 solid fractions). The results of MD simulations will allow us to propose scaling laws for the mechanical properties as a function of solid fraction and to evaluate the influence of ligament diameter on the mechanical properties.

The frameworks generated with Zeo++ were passed to a Python script that connects vertices (carbon atoms) by placing cylinders representing ligaments through linkers (oxygen atoms). The results of mesh generation are shown in Fig. 3, constructed from the framework shown in Fig. 1. The Marching Cubes algorithm, described in Ref. 20 and implemented as a Python library,²¹ was used to generate the triangulated mesh.

Figure 4 presents additional examples of generated structures. The generated structures are fully periodic, which is important for MD simulation of infinite-sized specimens.

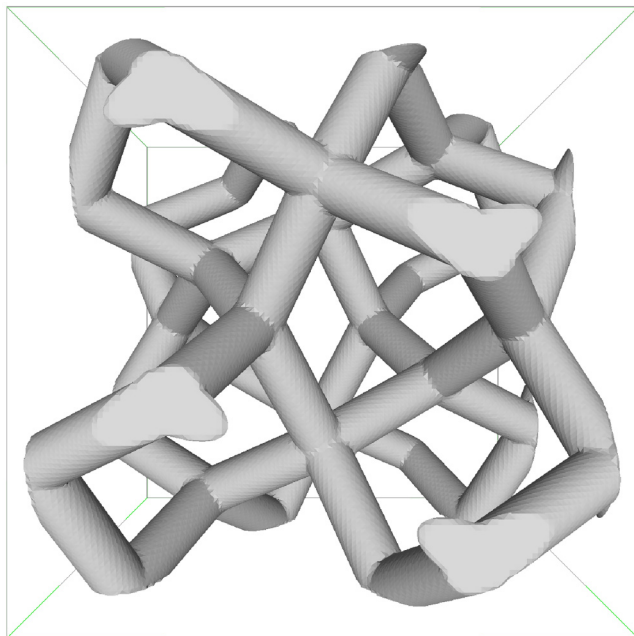


FIG. 3. *lcs* topology as a triangulated mesh.

The constructed *.stl* files were voxelized into 3D binary grids, where 0s and 1s indicate whether the point is outside or inside the mesh surface. The binary arrays were filled with an aluminum FCC lattice with a lattice constant 0.4046 nm at every point labeled as 1. The obtained *.imp* simulation input files store the total number of atoms, atom types, dimensions of the simulation box, and coordinates of each individual atom.

B. Stochastic NPA structures

Method proposed by Soyarslan *et al.*²² was used to generate stochastic nanoporous metal structures. This method involves the superpositioning of sinusoidal waves with fixed wavelengths but random phases and directions. By varying the input parameters, we generated 75 stochastic nanoporous structures in total. These structures are represented by binary 3D arrays with a resolution of $100 \times 100 \times 100$ points, where 0s and 1s are void and solid parts of the structure, respectively. The generated structures cover a solid fraction range of approximately 20%–80%. For some highly porous structures, we detected disconnected solid parts to remove them during post-processing. The 3D arrays were then used to construct *.imp* data files by placing the FCC lattice of Al atoms at each solid point in the array. With a lattice constant of 0.4046 nm, the size of all the stochastic structures was $40.46 \times 40.46 \times 40.46$ nm. The simulation domain size of this study is approximately 2^3 larger than that of a reference MD study on nanoporous aluminum.¹¹ While periodic boundary conditions, applied in both the reference and this study, are designed to avoid size dependence, this increase in domain size was introduced to capture potential size effects that might still be present. Figure 5 shows examples of generated stochastic NPA structures, visualization of all simulated systems was done with open-source OVITO software.²³ Applying Common Neighborhood Analysis, it was confirmed that the atoms at the structure boundaries were also in FCC lattice, indicating that the generated structures are indeed periodic in all three directions.

C. Molecular dynamics simulations

Molecular dynamics simulations were performed with the open-source code LAMMPS.²⁴ Embedded-atom method (EAM) potential²⁵ was used to describe the interaction between Al atoms during the simulations. This method is widely used to compute pairwise atomic interactions for metals and alloys. The total energy E_i of an atom i is calculated as

$$E_i = F_\alpha \left(\sum_{j \neq i} \rho_\beta(r_{ij}) \right) + \frac{1}{2} \sum_{j \neq i} \phi_{\alpha\beta}(r_{ij}), \quad (5)$$

where F_α is the embedding energy as a function of the local electron density ρ , $\phi_{\alpha\beta}(r_{ij})$ is a pair potential interaction, α and β are the element types of atoms i and j , respectively.

All the orthogonal simulation systems were periodic in three spatial directions to reproduce bulk behavior and avoid the effects of disconnected ligaments. For all stochastic NPA structures, the size of the simulation box was the same—40.46 nm, while for topologically defined structures, the box dimensions varied. The time step was set to 0.003 ps or 3 fs.

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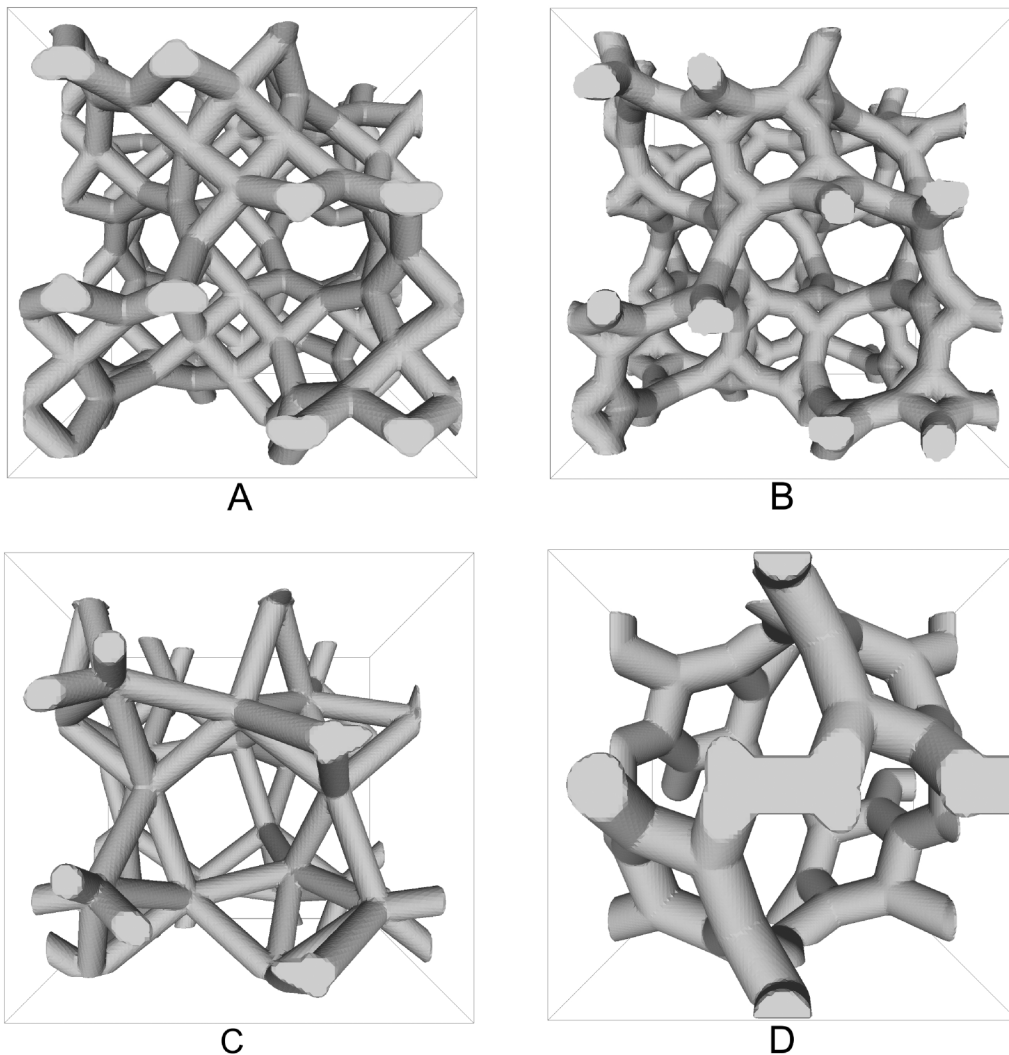


FIG. 4. Examples of topologies generated as a triangulated mesh: (a) *gie*, (b) *ana*, (c) *sxe*, and (d) *srd*.

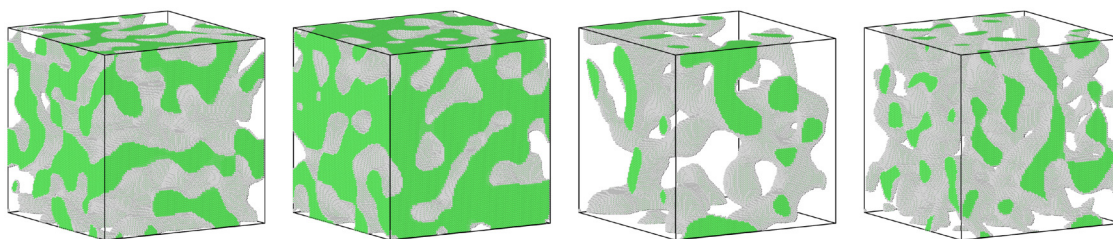


FIG. 5. Examples of generated stochastic NPA structures (atoms in green are in FCC lattice).

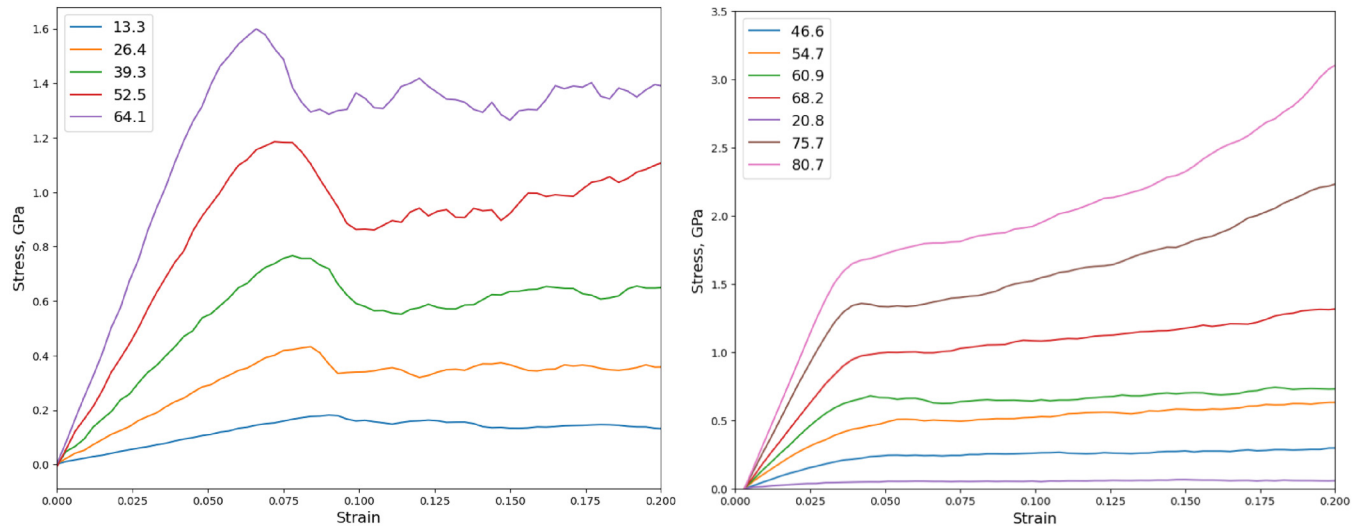


FIG. 6. Stress-strain curves of *dia* topology with 5 nm ligaments (left) and stochastic structures (right) with varying solid fraction (%).

The following simulation workflow was implemented for all stochastic and defined specimens. First, an energy minimization procedure was performed for the system with the Hessian-free truncated Newton algorithm.²⁶ This algorithm mostly behaves as the CG (Conjugate Gradient) algorithm,²⁷ but in our case, CG showed poor minimization results. Both of these algorithms are implemented as functions within the LAMMPS code. After minimization, the system was imposed to NVE equilibration for 90 ps at 300 K with a Langevin thermostat.^{28–30} This allowed the energy to redistribute among atoms and the system to reach thermal

equilibrium. Following the thermal equilibration, the system underwent the NPT ensemble for 90 ps. Finally, after minimization and NVE/NPT equilibration, the system was imposed to uniaxial compressive strain with the NVE and strain rate of 10^9 s^{-1} . Although the applied strain rate is quite high, this is common practice in MD simulations due to computational limitations.^{10,31,32} The strain rate sensitivity is discussed in Sec. III A. The resulting true stress of a system was calculated based on the stress components of each individual atom. As a final simulation result, we obtained strain-stress diagrams to calculate yield stress and Young's modulus. A full LAMMPS code can be found in the [supplementary material](#).

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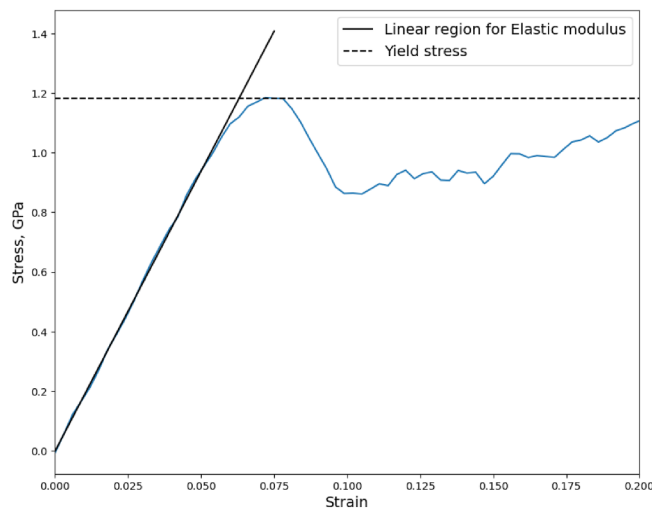


FIG. 7. Example of the mechanical properties calculation.

III. RESULTS AND DISCUSSION

A. MD simulations results

In total, this study involved 405 simulations: 75 simulations of stochastic nanoporous aluminum structures and 330 simulations of

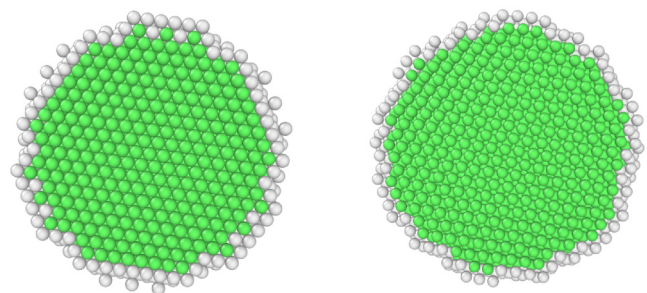


FIG. 8. Cross section of 5 nm ligament before (left) and after (right) energy minimization (atoms in green are in FCC lattice).

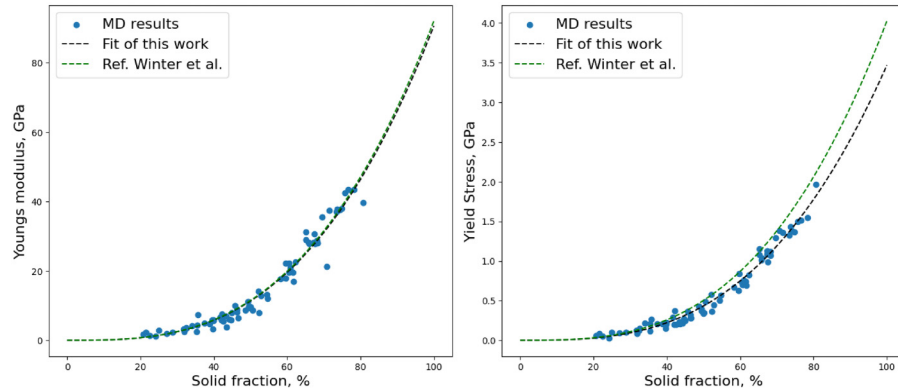


FIG. 9. Simulation results and scaling laws for mechanical properties of stochastic NPA structures.

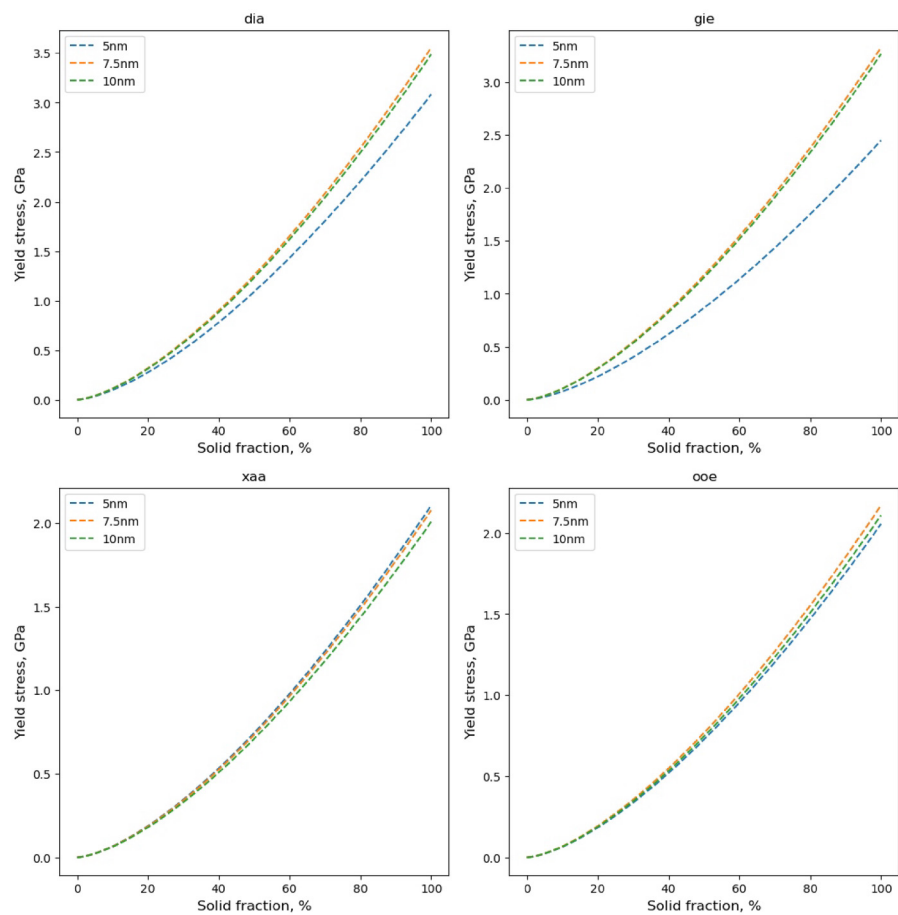


FIG. 10. Examples of differences in yield stress scaling laws.

topologically defined structures—22 topologies each with 3 ligament diameters and 5 solid fraction values. During the simulations, the necessary parameters of the system were saved from LAMMPS for analysis. Examples of the obtained stress–strain curves for *dia* topology and stochastic structures are shown in Fig. 6. Mechanical properties were calculated from each stress–strain diagram: Young's modulus and yield stress as shown in Fig. 7.

As the structures of this work are constructed by removing porous regions from a bulk aluminum, the ligaments might be affected by the sharp edges of their surface after relaxation. To ensure that this is not the case, we monitored the generated

structures before and after relaxation. It was confirmed that relaxed structures remained free of internal defects, such as stacking faults and dislocations, which were not present both before and after energy minimization (Fig. 8). This observation is supported by previous studies on nanoporous metals,¹⁹ the authors concluded that the diameter of 5–6 nm is sufficient for ligaments to be free from defects after relaxation. As all of the structures in this study were built with a ligament diameter of 5, 7.5, or 10 nm, the simulation results are not affected by the method of structure generation.

The strain rate sensitivity was also investigated by performing simulations with strain rates twice as high ($2 \times 10^9 \text{ s}^{-1}$) and twice

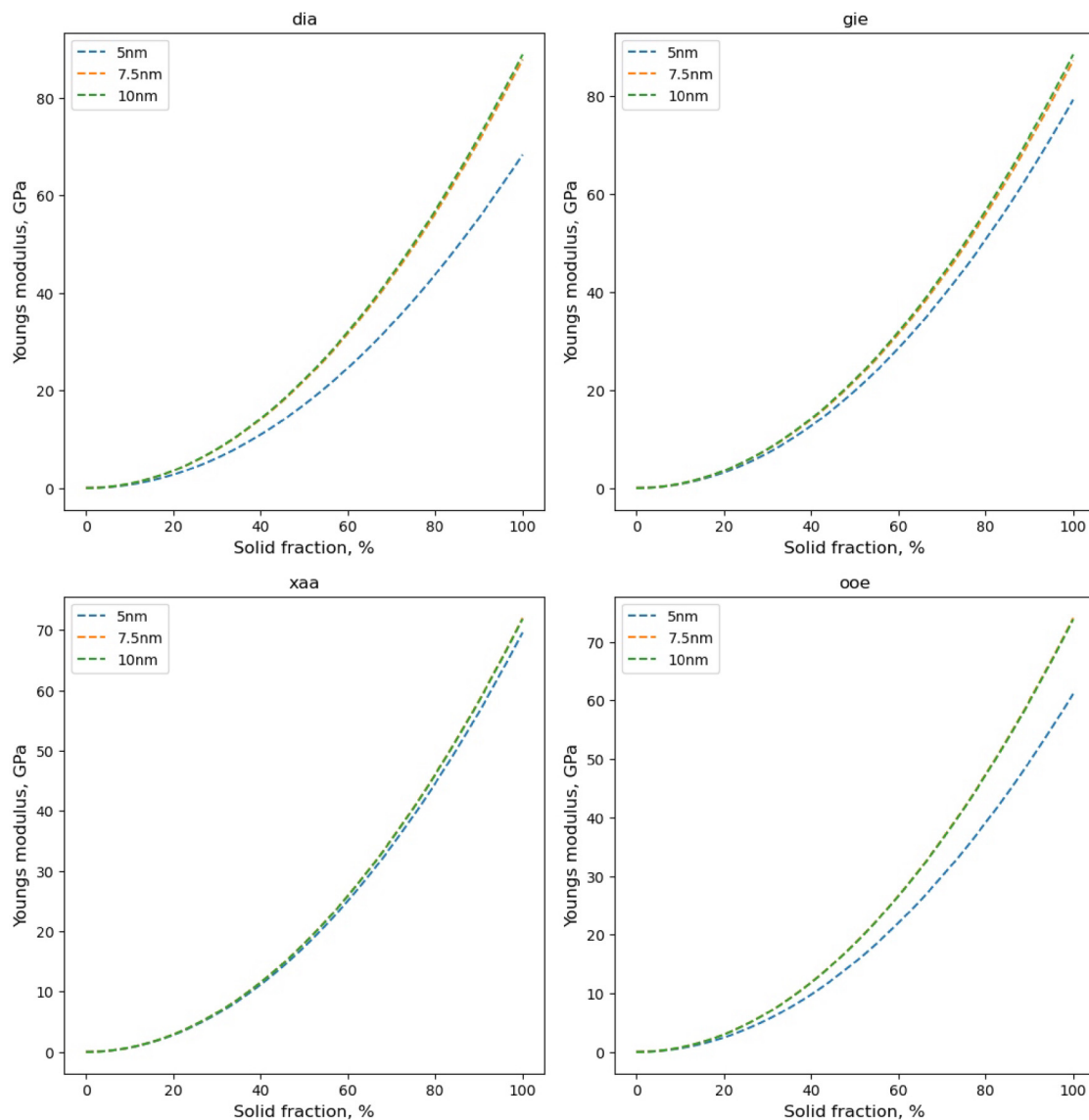


FIG. 11. Examples of differences in Young's modulus scaling laws.

TABLE I. Percentage differences ($\Delta\%$) in yield stress and Young's modulus prefactors for varying ligament diameters.

Topology	ΔC_σ (10 nm, 5 nm)	ΔC_σ (10 nm, 7.5 nm)	ΔC_E (10 nm, 5 nm)	ΔC_E (10 nm, 7.5 nm)
dia	11.59%	1.97%	23.11%	1.14%
lcs	51.69%	1.97%	29.52%	1.14%
urh	50.17%	2.00%	34.45%	1.22%
gie	24.96%	1.08%	10.44%	1.36%
ukx	30.44%	1.67%	28.39%	1.02%
bcv	35.37%	1.14%	10.69%	2.53%
ana	3.58%	1.16%	1.74%	2.54%
lcs-f	9.45%	1.33%	9.30%	2.53%
pbg	24.92%	0.76%	19.75%	2.57%
pbz-l	29.13%	0.04%	16.55%	2.97%
xaa	4.93%	3.45%	3.10%	0.22%
srd	11.13%	3.46%	1.23%	0.22%
ttc	28.97%	3.62%	20.50%	0.24%
ooe	2.52%	2.91%	17.17%	0.32%
mct	10.67%	2.83%	10.96%	0.37%
kgq	30.15%	0.25%	29.01%	1.54%
sxg	26.70%	0.28%	29.05%	1.53%
sxn	22.29%	0.19%	25.81%	1.59%
lcy	36.43%	0.08%	31.11%	1.85%
smq	24.55%	1.03%	23.26%	2.57%
sxe	14.76%	1.57%	3.83%	2.63%
sxf	15.59%	1.49%	18.97%	2.61%

TABLE II. Prefactors of yield stress scaling law for tested topologies.

Topology	$C_\sigma \times 10^3$	Topology	$C_\sigma \times 10^3$	Topology	$C_\sigma \times 10^3$	Topology	$C_\sigma \times 10^3$
dia	3.488	sxn	2.634	sxe	2.463	lcy	2.203
bcv	3.226	ana	2.628	ukx	2.441	lcs	2.151
ttc	3.091	srd	2.623	sxf	2.435	pbg	2.105
mct	3.036	kgq	2.594	sxg	2.434	urh	2.007
gie	2.828	lcs-f	2.497	xaa	2.397		
pbz-l	2.811	ooe	2.491	smq	2.391		

TABLE III. Prefactors of elastic modulus scaling law for tested topologies.

Topology	$C_E \times 10^3$	Topology	$C_E \times 10^3$	Topology	$C_E \times 10^3$	Topology	$C_E \times 10^3$
mct	10.959	dia	8.893	xaa	8.496	lcy	7.767
bcv	10.623	lcs-f	8.838	sxg	8.464	lcs	7.576
pbz-l	10.378	sxn	8.805	ooe	7.996	ttc	7.552
gie	9.798	kgq	8.781	ukx	7.991	urh	7.181
sxe	9.622	smq	8.634	sxf	7.890		
ana	9.533	srd	8.612	pbg	7.787		

as low ($5 \times 10^8 \text{ s}^{-1}$) as the original rate for *dia* and *lcs* topologies. These simulations showed that, while the higher strain rate noticeably changed the mechanical response, such a high strain rate ($2 \times 10^9 \text{ s}^{-1}$) is not realistic for compression/tension and is more typical for shock wave simulations. Therefore, we simulated an additional four topologies with lower strain rates and observed minimal deviations in scaling laws. This consistency confirmed that the original strain rate provided reliable mechanical properties.

B. Scaling laws for stochastic NPA structures

Figure 9 shows that the MD-computed values for both Young's modulus and yield stress of stochastic aluminum structures follow a power law relationship with the solid fraction. As a reference, power laws reported in Ref. 11 are also plotted as green lines. The results for the elastic modulus are nearly identical to the reference, while there is a slight deviation observed for the yield stress. This difference may arise due to the following factors: the power law in the reference was proposed by fitting 6 data points, while we fitted 75; the simulations of the reference were performed for tensile tests, while we simulated compression tests. Taking into account these differences, it was concluded that MD simulation results of this study are consistent with the reference, and modified scaling laws were proposed for Young's modulus and yield stress, respectively,

$$E = 90.66\rho^3 \quad (\text{GPa}), \quad (6)$$

$$\sigma_s = 3.469\rho^3 \quad (\text{GPa}). \quad (7)$$

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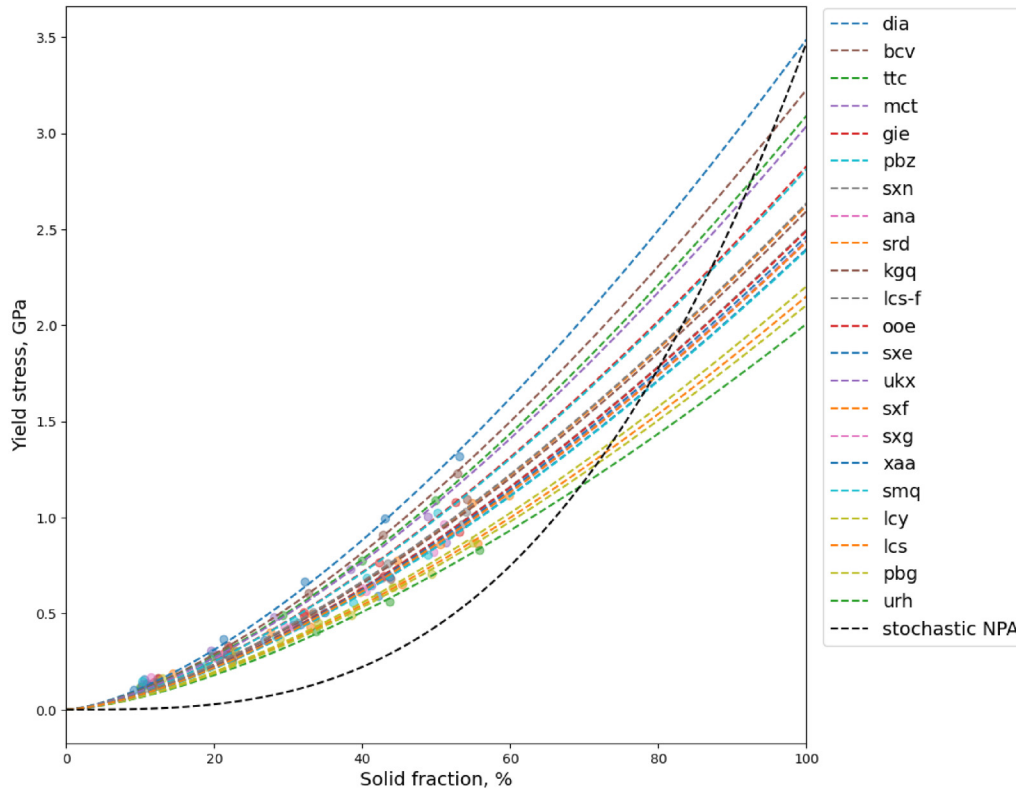


FIG. 12. Scaling curves of yield stress for tested topologies and stochastic NPA.

Experimental validation of the proposed scaling laws remains limited, as most of the experimental studies on nanoporous metals focus on gold,^{33,34} aluminum oxides,^{35,36} or aluminum alloys.³⁷ Nevertheless, a recent study by Pinna *et al.*³⁸ reported a fabrication of nanoporous aluminum with Vapor-Phase Dealloying (VPD) from $Al_{20}Zn_{80}$ nanocomposite. The authors measured the elastic modulus of the fabricated NPA with nanoindentation and obtained a value of 18.3 ± 1.4 GPa. When we apply Eq. (6) with a solid fraction of 0.6 (as estimated from the pore volume by the authors), the predicted Young's modulus is 19.58 GPa, showing a good agreement with the experiments. Although the comparison is based on a single available datapoint, it suggests that computationally derived scaling laws capture key trends in the mechanical properties of nanoporous aluminum.

C. Scaling laws for topologically defined structures

As previously described, 15 copies were generated per topology: 3 different ligament diameters with 5 different values of as-designed solid fraction (0.1, 0.2, 0.3, 0.4, 0.5), which were later recalculated from *.Imp* files for the as-built solid fraction. The mechanical properties of these five copies with the same topology, varying in solid fraction, were fitted to derive scaling laws for yield stress and elastic modulus. The power laws for topologically defined structures are expressed as follows:

$$\sigma = C_{\sigma} \rho^{1.5} \quad (\text{GPa}), \quad (8)$$

$$E = C_E \rho^2 \quad (\text{GPa}), \quad (9)$$

where C_{σ} and C_E are prefactors for yield stress and Young's modulus, respectively.

To investigate the effect of ligaments diameter on scaling laws, we first derived three scaling laws per topology, i.e., each one fitted with a different ligament diameter. Examples of how the scaling laws for yield stress and Young's modulus deviate with changing ligament diameter are shown in Figs. 10 and 11, while deviations for the studied topologies are summarized in Table I. Comparing these deviation values, we observed that scaling laws remain nearly consistent for 7.5 and 10 nm diameters, while showing noticeable deviations at 5 nm. This suggests the existence of a "critical" ligament diameter between 5 and 7.5 nm, where the behavior of porous structures changes due to a higher surface-to-volume ratio with higher surface effects. Therefore, the scaling laws proposed in this study are based on simulation results for structures with a ligament diameter of 10 nm.

The prefactors of the scaling laws (8) and (9) for the tested topologies are presented in Table II and Table III, respectively. The highest scaling law prefactors for yield stress were observed in diamond topology and its four-connected analog *bcv*, followed by

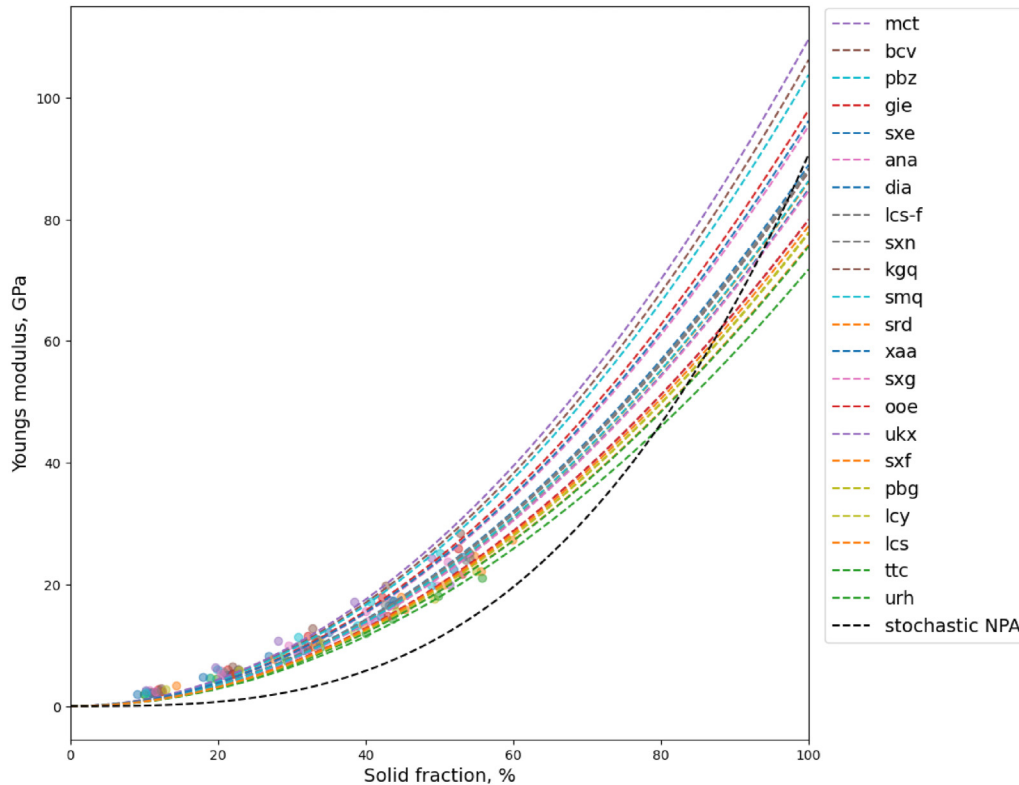


FIG. 13. Scaling curves of Young's modulus for tested topologies and stochastic NPA.

three-connected topology *ttc*. For the elastic modulus, the highest prefactors were found in three-connected topologies *mct* and *pbz*-1 and four-connected *bcv*. Additionally, the topology had a stronger effect on the mechanical properties at higher values of solid fraction. For example, at a solid fraction of 60%, the yield stress of *dia* topology is 1.75 times greater than that of the *urh* topology, while the difference is smaller for the elastic modulus.

To compare the mechanical behavior of stochastic and defined structures, the scaling laws for tested topologies were plotted in one figure with the scaling laws of stochastic NPA. As shown in Figs. 12 and 13, up to approximately 70% solid fraction, every tested topology (colored lines) has a higher yield stress and Young's modulus than stochastic structures (black line) at the same solid fraction. It should be noted that all the fit lines for the defined structures as well as the power law of stochastic structures must have the same value for the bulk material (100% solid fraction). However, since the focus of this study was not on those extreme cases, the fit lines are shown across the whole solid fraction range but are considered valid only up to the highest simulated value.

Previous studies on porous structures^{39,40} have reported that the differences in mechanical properties between stochastic and architected structures come from their deformation mechanisms. Stochastic structures usually show stretch-dominated deformation

behavior, while ordered structures primarily deform by stretching of ligaments. These mechanisms are also observed from the simulated stress-strain curves (Fig. 6) when compared with curves and behavior reported in the literature.⁴¹ For instance, simulated stochastic structures do not exhibit any post-yield softening and show a flat stress plateau region, as their ligaments are deformed simultaneously. On the other hand, topologically defined structures exhibit clear post-yield softening with peaks on their stress plateau, corresponding to the failure of structural layers.

To explain the variation between different topologies and therefore their scaling laws, scaled genus density was computed as follows:

$$\bar{g}_v = \frac{g}{V} \times S_v^{-3}, \quad (10)$$

where g is the genus of the structure, V is the total volume of the system, and S_v is the specific surface (surface per volume). The genus of the closed triangulated surface can be calculated as follows:

$$g = 1 - \frac{V - E + F}{2}, \quad (11)$$

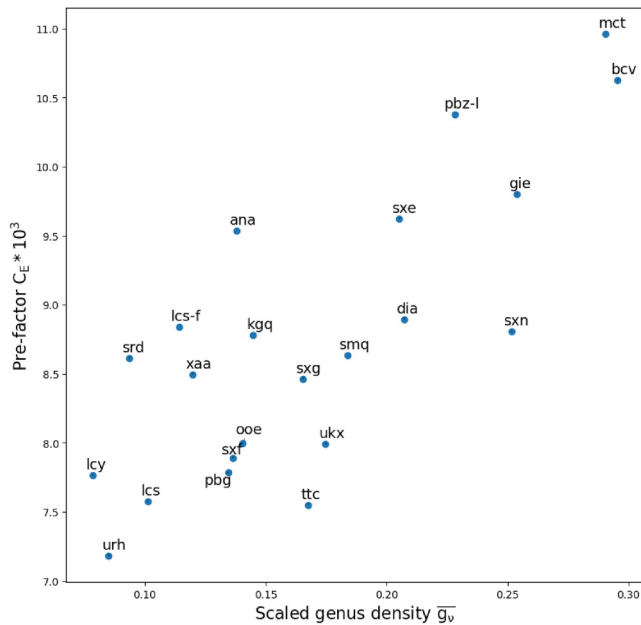


FIG. 14. Scaled genus density vs scaling law prefactor for Young's modulus.

where V , E , F are the number of vertices, edges, and faces of a triangulated surface. Figures 14 and 15 show the dependence of the prefactors C_E and C_σ , respectively, on the scaled genus density for the 22 tested topologies. As can be seen, the prefactors generally

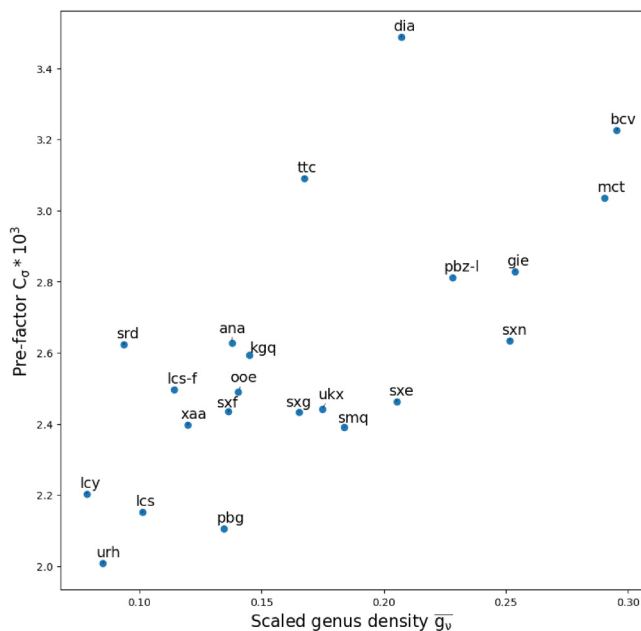


FIG. 15. Scaled genus density vs scaling law prefactor for yield stress.

increase with increasing scaled genus density. The Pearson correlation coefficient is 0.7854 ($p\text{-value} = 1.49 \times 10^{-5}$) for elastic modulus and 0.708 ($p\text{-value} = 2.24 \times 10^{-4}$) for yield stress. For example, the three “weakest” topologies (*pbz-l*, *lcs*, and *urh*) exhibit the lowest scaled genus density, while the “strongest” (*dia*, *bcv*, and *mct*) have higher scaled genus density values. The same trend is observed for Young's modulus, with the prefactors increasing together with the scaled genus density. Although the genus and prefactors are not perfectly correlated, this relation provides valuable insight into how connectivity and topology influence the mechanical properties of porous structures.

Scaled genus density was also calculated for stochastic NPA structures. However, no correlation with mechanical or structural properties was observed. In the work of Guillotte *et al.*,³² the authors reported a general trend of $\bar{g}_v$ increase with increasing ρ based on a few data points available. Overall, they suggested that scaled genus density in stochastic structures might not only depend on relative density. Therefore, no scaling laws were proposed for $\bar{g}_v$ due to insufficient data. A detailed description of $\bar{g}_v$ calculation can be found in the [supplementary material](#).

IV. CONCLUSION

In this study, we proposed a new method for generating chemistry-inspired porous structures with defined topology and evaluated their mechanical properties using Molecular Dynamics simulations. This approach is based on collecting data from the RCSR database, assembling a periodic unit cell with Zeo++, and generating a triangulated mesh. It provides a flexible workflow that can be adapted to other applications, including 3D printing and finite element simulations. The method can be adapted as a foundation for future studies on properties, design, and optimization of the architected materials.

Molecular dynamics simulations revealed that the mechanical properties of nanoporous aluminum are sensitive to ligament diameter. A noticeable deviation in the scaling laws was observed at a diameter of 5 nm due to increased surface effects. For larger values of diameter (7.5 and 10 nm), the mechanical properties and their scaling laws were consistent, suggesting that the critical ligament diameter can be found below 7.5 nm.

By comparing stochastic NPA structures with architected ones, it was found that topologies demonstrate superior yield stress and Young's modulus for the equivalent solid fraction. Among the tested topologies, the highest yield stress was observed in four-connected *dia* and *bcv* and three-connected *ttc*, while three-connected *mct* and *pbz-l* with *bcv* achieved the highest elastic modulus. The structure-property relationship was summarized by topology-dependent scaling laws, demonstrating the great influence of topology on the mechanical performance as each of the structures has a distinct scaling law coefficient.

Furthermore, a positive correlation between scaled genus density and scaling laws coefficients was observed for topologically defined structures. It suggests that the connectivity of the ligament network has a key role in defining the mechanical behavior of the whole porous structure. The findings of this study provide insights in the understanding of how topology influences strength and stiffness of architected materials and provide a methodological workflow for their generation and mechanical evaluation.

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SUPPLEMENTARY MATERIAL

See the [supplementary material](#) for the information about the scaled genus density calculation and the full LAMMPS code used for MD simulations.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Sergei Zorkaltsev: Conceptualization (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Software (equal); Visualization (equal); Writing – original draft (equal). **Maciej Haranczyk:** Conceptualization (equal); Funding acquisition (lead); Project administration (lead); Resources (lead); Supervision (lead); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding authors upon reasonable request.

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